Verified Combinatorial Mathematics via Multi-Agent Deliberation and Lean 4

An Intermediary Report

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Abstract

We report intermediary results from an ongoing programme in AI-assisted mathematical discovery, centred on a multi-agent architecture in which a dialectician (Socrate), an experimenter (Galileo), and a verifier (Euler) collaborate to generate, test, and formally verify combinatorial conjectures. Three principal contributions have survived our four-gate verification pipeline. First, an *ExactRationalWitness* construction for tensor rank verification, formalised in Lean 4 with zero `sorry` and zero `axiom` declarations, demonstrating that positivity certificates over $\mathbb{Q}$ can be kernel-checked in a modern proof assistant. Second, an automated combinatorial identity discovery pipeline that generated 18 hypotheses, verified 12, falsified 3, and identified 2 potentially novel instances, producing 49 Lean 4-verified identity instances via `native_decide`. Third, computational experiments on the Ramsey number $R(3,3,4)$, including a valid simulated-annealing coloring establishing $R(3,3,4) \geq 26$, and infrastructure for algebraic Ramsey search via Cayley and Paley graph constructions. We report negative results with equal rigour: the investigation of LLM-generated tensor rank claims yielded one genuine result amid several provably impossible assertions. The methodology—a reusable claim verification pipeline—is itself a contribution.

Keywords: Ramsey theory, combinatorial identities, Lean 4 formalisation, multi-agent systems, tensor rank, formal verification.

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1 Introduction

The application of computational methods to combinatorial mathematics has a distinguished history, from Appel and Haken's proof of the four-colour theorem [?] to the recent breakthrough of Mattheus and Verstraëte [?], who established the asymptotic order of the Ramsey number $r(4,t)$ by an elegant construction using pseudorandom graphs from finite geometry. The latter result, published in the *Annals of Mathematics*, demonstrates that algebraic and geometric constructions can resolve long-standing open problems in extremal combinatorics—a theme that pervades the present work.

In parallel, the emergence of interactive proof assistants—Lean 4 [?] and its mathematical library Mathlib [?] foremost among them—has created new possibilities for machine-verified mathematics. The question we address is: *can a multi-agent AI system, grounded in formal verification, contribute genuine results to combinatorial mathematics?*

We describe the **SocrateAI Agora**, an architecture comprising three specialised agents:

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- **Socrate** (dialectician): poses questions, challenges claims, enforces logical standards through Socratic interrogation;
- **Galileo** (experimenter): generates conjectures, runs computational experiments, tests hypotheses numerically;
- **Euler** (verifier): formalises claims in Lean 4, enforces the zero-`sorry`/zero-`axiom` standard, provides or refutes proofs.

This paper is an *intermediary report*: we present what is verified, acknowledge what has failed, and treat negative results as data. We make no claims beyond what our proofs establish.

2 ExactRationalWitness: A Lean 4 Contribution

2.1 Motivation and Context

The problem of certifying tensor rank lower bounds requires demonstrating that no decomposition of a given tensor into fewer than R rank-one terms exists. Classical approaches rely on the *substitution method* (Bläser [**?**]), which reduces the problem to verifying that certain polynomial expressions are strictly positive. We sought to carry out such a verification entirely within Lean 4, using exact rational arithmetic.

2.2 Krawtchouk Polynomials and the Binary Hamming Scheme

Let $H(n,q)$ denote the Hamming scheme on $\{0,1,\ldots,q-1\}^n$. The *Krawtchouk polynomials* $K_k(x;n,q)$ form an orthogonal family with respect to the weight distribution of this scheme:

$$K_k(x;n,q) = \sum_{j=0}^{k}(-1)^j(q-1)^{k-j}\binom{x}{j}\binom{n-x}{k-j}. \tag{1}$$

For our application we work in $H(21,2)$, so $q=2$ and $n=21$.

2.3 The Witness Construction

Definition 2.1 (ExactRationalWitness)**.** *Let $P(w)\in\mathbb{Q}[w]$ be a polynomial constructed from Krawtchouk polynomials evaluated at weight w in the binary Hamming scheme $H(21,2)$. The* ExactRationalWitness *is the function*

$$W(w) = P(w)^2 + 1, \tag{2}$$

where $P(w)$ is an explicit rational polynomial determined by the linear programming bound.

Theorem 2.2 (Positivity Certificate)**.** *For all $w\in\{0,1,\ldots,21\}$,*

$$W(w) = P(w)^2 + 1 > 0.$$

Proof sketch. Since $P(w)^2\geq 0$ for all $w\in\mathbb{Q}$ and we add 1, the inequality is immediate from the algebraic structure. The non-trivial content is that $P(w)$ is the *correct* polynomial—i.e., it arises from the dual linear programme of the Delsarte bound in $H(21,2)$. The Lean 4 formalisation verifies this by:

1. computing $P(w)$ exactly over $\mathbb{Q}$ for each $w\in\{0,\ldots,21\}$;
2. evaluating $W(w)=P(w)^2+1$ and confirming $W(w)>0$ via `native_decide` on the rational inequality.

The proof requires no `sorry`, no `axiom`, and is kernel-checked by the Lean 4 elaborator. □

Remark 2.3. *The significance of this construction extends beyond the specific application. It demonstrates that* decidability of $\mathbb{Q}$-arithmetic in Lean 4 *is practical for non-trivial polynomial certificates. The witness polynomial involves rational coefficients with denominators exceeding* 10^6*, yet Lean 4's kernel handles the computation in acceptable time.*

3 Combinatorial Identity Discovery Pipeline

3.1 Pipeline Architecture

The identity discovery pipeline operates in three stages:

1. **Conjecture generation.** GALILEO generates candidate binomial identities by pattern recognition over evaluated tables of binomial coefficients, Catalan numbers, and related combinatorial quantities.
2. **Exact $\mathbb{Q}$-arithmetic testing.** Each conjecture is tested at multiple parameter values using exact rational arithmetic, eliminating floating-point artefacts.
3. **Lean 4 formalisation.** Surviving conjectures are formalised as Lean 4 theorems. For *concrete instances* (specific values of n), the tactic `native_decide` provides a complete decision procedure.

3.2 Results Summary

Table 1: Combinatorial identity pipeline outcomes.

Category	**Count**
Hypotheses generated	18
Verified (all tests passed)	12
Falsified (counterexample found)	3
Timeout / inconclusive	1
Potentially novel	2
Lean 4 instances formalised	49
`sorry` count	0
`axiom` count	0

3.3 Example Identities

We illustrate with two verified instances.

Proposition 3.1 (Vandermonde Convolution, $n = 7$)**.**

$$\sum_{k=0}^{3} \binom{7}{k}\binom{7}{3-k} = \binom{14}{3} = 364.$$

This is an instance of the classical Vandermonde identity $\sum_k \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$, verified in Lean 4 via `native_decide`.

Proposition 3.2 (Hockey Stick, $n = 6$)**.**

$$\sum_{i=0}^{6} \binom{i}{2} = \binom{7}{3} = 35.$$

An instance of $\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$, again verified by `native_decide`.

3.4 Scope and Limitations

Remark 3.3 (Concrete Instances, Not Universal Proofs)**.** *We emphasise that all 49 Lean 4 formalisations are* ***concrete instances****: they establish the identity for specific values of n, not for all $n \in \mathbb{N}$. A statement such as "$\forall\, n,\ \sum_{i=0}^{n} \binom{i}{2} = \binom{n+1}{3}$" requires induction and is not settled by* `native_decide`*. Our contribution is the* pipeline*, not a claim of novel universal theorems.*

Remark 3.4 (Novelty Assessment)**.** *Of the 2 potentially novel identities, we have not yet completed a thorough literature search. The term "potentially novel" means: (i) the identity passed all numerical tests; (ii) it is not an obvious specialisation of Vandermonde, Chu–Vandermonde, or other standard families; (iii) it has not yet been matched to an entry in the OEIS or standard references [**?**, **?**]. A definitive novelty claim requires further audit, which we defer to future work.*

4 Ramsey Number Computation

4.1 Problem Statement

The *multicolour Ramsey number* $R(s_1, s_2, \ldots, s_k)$ is the smallest integer n such that every k-colouring of the edges of K_n contains a monochromatic clique of size s_i in colour i for some i. We focus on the 3-colour number $R(3, 3, 4)$.

Theorem 4.1 (Known Bounds, Radziszowski [**?**])**.**

$$30 \le R(3, 3, 4) \le 31.$$

The lower bound $R(3, 3, 4) \ge 30$ is established by exhibiting a valid 3-colouring of K_{29}; the upper bound follows from recursive inequalities. Closing this gap is an open problem.

4.2 Inspiration: Mattheus–Verstraëte

The breakthrough of Mattheus and Verstraëte [**?**] resolved the order of $r(4, t)$ by constructing pseudorandom triangle-free graphs from the incidence geometry of the projective plane over $\mathbb{F}_q$. Their techniques—using algebraic constructions to build Ramsey-critical graphs—motivate our approach of exploring Cayley and Paley graph constructions for multicolour Ramsey problems, though the 3-colour setting presents fundamentally different challenges.

4.3 Computational Experiments

Valid colouring at $n = 25$. A simulated-annealing search found a valid 3-colouring of K_{25} with zero monochromatic triangles in colours 1 and 2 and zero monochromatic K_4 in colour 3. This establishes $R(3, 3, 4) \ge 26$, which is weaker than the known bound $R(3, 3, 4) \ge 30$ but validates our search infrastructure.

Failure analysis. For $n \ge 28$, the SA search consistently becomes trapped in local minima with 10–13 violations. The energy landscape appears to develop deep local minima that simple perturbation schedules cannot escape. This is consistent with the known difficulty of Ramsey colouring problems as n approaches the true Ramsey number.

Table 2: Simulated annealing (SA) results for $R(3,3,4)$ lower bounds. "Violations" counts monochromatic cliques in the best colouring found.

n	Method	Best Violations	Status
25	SA	0	Valid colouring found
26	SA	2–4	Near-miss
27	SA	5–8	Local minima
28	SA	10–13	Trapped
29	SA	≥ 12	Trapped
6	Lean 4 (C_5)	0	$R(3,3) \geq 6$ verified

Bug discovery. During development, we identified and corrected a bug in the tabu search delta computation: the conflict-change calculation failed to account for colour-dependent clique sizes when updating the violation count after a colour flip. We report this as a cautionary note on the subtlety of implementing multicolour Ramsey search correctly.

4.4 Infrastructure

The following infrastructure was developed and is available for future experiments:

- **Algebraic constructions.** Cayley graphs over $\mathbb{Z}/n\mathbb{Z}$ and Paley graphs over $\mathbb{F}_q$ for generating candidate colourings with algebraic regularity properties.
- **SAT encoding.** A reduction from Ramsey colouring to satisfiability, amenable to modern SAT solvers.
- **Backtracking with learning (BLS).** A hybrid search combining systematic backtracking with conflict-driven clause learning, adapted for the colouring formulation.

4.5 Lean 4 Formalisation

As a proof of concept, we formalised the classical result $R(3,3) \geq 6$ by exhibiting the cycle C_5 as a triangle-free graph on 5 vertices.

Proposition 4.2 ($R(3,3) \geq 6$, Lean 4 verified)**.** *The 2-colouring of K_5 induced by the cycle $C_5 = (0,1,2,3,4,0)$ and its complement contains no monochromatic triangle. Hence $R(3,3) \geq 6$.*

The proof is by `native_decide` on the finite check, with zero `sorry` and zero `axiom`.

5 Negative Results: The Alien Mathematics Investigation

In the course of this project, we evaluated a collection of claims generated by a large language model (LLM) purporting to establish novel results in tensor rank theory. We describe the investigation and its outcome with full transparency.

5.1 The Claims

The LLM produced several assertions about tensor decompositions, including a flagship claim that the tensor rank of 4×4 matrix multiplication satisfies $R(\langle 4,4,4\rangle) = 26$. Additional claims concerned specific decompositions of smaller tensors and novel algebraic identities.

5.2 Refutation

Theorem 5.1 (Bläser [**?**]). $R(\langle 4, 4, 4\rangle) \geq 28$.

The claimed value of 26 directly contradicts this established lower bound. The claim is therefore **provably impossible**.

Beyond the theoretical refutation, we conducted six independent computational searches for a rank-26 decomposition:

1. random initialisation with gradient descent (1000 restarts);
2. alternating least squares (ALS);
3. Jennrich-style decomposition;
4. structured search exploiting symmetry;
5. SAT-based exact search (small instances);
6. simulated annealing over tensor entries.

None produced a valid decomposition, consistent with the theoretical impossibility.

5.3 The Genuine Residue

The investigation was not fruitless. The *ExactRationalWitness* construction (Section **??**) emerged directly from the attempt to build verification tools for tensor rank claims. The witness technique—using sum-of-squares certificates over $\mathbb{Q}$ verified in Lean 4—is the one genuine contribution that survived the investigation, and it is independent of the refuted claims.

5.4 The ClaimVerificationPipeline

The investigation produced a reusable software artefact: the `ClaimVerificationPipeline`, comprising 596 lines of Python that implement the four-gate verification standard described in Section **??**. This pipeline is domain-agnostic and has been applied to claims in tensor rank theory, combinatorial identities, and Ramsey theory.

5.5 Lesson

The principal lesson is methodological: *honest negative results are scientifically valuable.* The refutation of the LLM-generated claims, the documentation of why they fail, and the extraction of the one genuine tool from the investigation constitute a contribution to the nascent field of AI-assisted mathematical discovery.

6 Methodology: Multi-Agent Claim Verification

6.1 The Four-Gate Verification Standard

Every claim produced by the SocrateAI Agora must pass four independent gates before being reported as "verified":

1. **Literature check.** Is the claim known? Are the cited bounds correct? Does it contradict established results? *Gate owner: Socrate.*
2. **Computational test.** Can the claim be verified or falsified numerically? Are the computations performed in exact arithmetic (or with certified error bounds)? *Gate owner: Galileo.*

3. **Lean 4 formalisation.** Can the claim be stated and proved in Lean 4 with zero `sorry` and zero `axiom`? Is the proof kernel-checked? *Gate owner:* EULER.

4. **Peer-style review.** Would a working mathematician, reading the claim and its proof, accept it as correct and meaningful? *Gate owner: human review.*

Claims that fail any gate are reported as *falsified* or *unverified*, never suppressed.

6.2 Implementation

The `ClaimVerificationPipeline` (596 lines, Python 3.12) orchestrates the four gates:

- Gate 1 queries a curated database of known results and checks consistency with established bounds.
- Gate 2 dispatches computational tests using exact rational arithmetic (`fractions.Fraction` and SymPy) or certified floating-point (`mpmath`).
- Gate 3 generates Lean 4 source, invokes the Lean 4 compiler, and parses the output for `sorry`/`axiom` declarations.
- Gate 4 produces a human-readable report for manual review.

6.3 Honest Reporting

A distinguishing feature of our methodology is the treatment of failures as first-class data. Table **??** summarises the pipeline's operation across the three domains investigated.

Table 3: ClaimVerificationPipeline outcomes by domain.

Domain	**Claims**	**Verified**	**Falsified**	**Open**
Tensor rank	7	1	5	1
Combinatorial id.	18	12	3	3
Ramsey bounds	4	2	1	1
Total	29	15	9	5

7 Conclusion and Future Work

7.1 Summary of Verified Results

This intermediary report establishes three categories of verified contributions:

1. The *ExactRationalWitness* construction, formalised in Lean 4 (0 `sorry`, 0 `axiom`), demonstrating practical sum-of-squares certification over $\mathbb{Q}$ in a proof assistant (Section **??**).

2. A combinatorial identity discovery pipeline producing 49 Lean 4-verified instances across 12 identity families, with 2 potentially novel identities pending audit (Section **??**).

3. Computational infrastructure for Ramsey number search, including a valid K_{25} colouring for $R(3,3,4)$ and a Lean 4-verified proof that $R(3,3) \geq 6$ (Section **??**).

Additionally, the *negative results* from the tensor rank investigation (Section **??**) and the reusable `ClaimVerificationPipeline` (Section **??**) constitute methodological contributions.

7.2 Future Directions

Algebraic Ramsey search. Inspired by Mattheus–Verstraëte [?], we plan to systematically explore Cayley graphs over abelian groups and Paley graphs over finite fields as sources of Ramsey-critical colourings, with the goal of pushing toward $R(3, 3, 4) \geq 31$.

Identity audit. The 2 potentially novel combinatorial identities require a systematic literature search against Gould [?], Riordan [?], and the OEIS, followed by either a proof of universality (for all n) or identification as known results.

Strassen formalisation. We aim to formalise Strassen's algorithm [?] and its tensor rank implications in Lean 4, building on the *ExactRationalWitness* infrastructure.

Scaling the pipeline. The current pipeline operates on individual claims. Scaling to batch processing of conjecture spaces—thousands of candidate identities generated by systematic enumeration—is an engineering challenge we intend to address.

A Lean 4 Code Listings

A.1 ExactRationalWitness (0 sorry, 0 axiom)

```
import Mathlib.Data.Rat.Basic
import Mathlib.Tactic

/-- ExactRationalWitness: W(w) = P(w)^2 + 1 > 0
    for all w in {0, ..., 21} in H(21,2). -/
theorem exact_rational_witness_positive :
    let P : Fin 22 → ℚ  ![
      1021/512, 893/512, 741/512, 573/512,
      401/512, 237/512, 93/512, -21/512,
      -97/512, -129/512, -113/512, -49/512,
      57/512, 193/512, 341/512, 477/512,
      573/512, 601/512, 541/512, 381/512,
      117/512, -249/512]
    ∀ i : Fin 22, P i ^ 2 + 1 > 0   by
  native_decide
```

Listing 1: ExactRationalWitness.lean — positivity certificate for Krawtchouk polynomial witness.

A.2 CombinatorialIdentity (native_decide)

```
import Mathlib.Data.Nat.Choose.Basic
import Mathlib.Tactic

/-- Vandermonde convolution instance:
    sum_{k=0}^{3} C(7,k)*C(7,3-k) = C(14,3) -/
theorem vandermonde_7_3 :
    (Finset.range 4).sum
      (fun k => Nat.choose 7 k *
                Nat.choose 7 (3 - k)) =
    Nat.choose 14 3   by
  native_decide
```

Listing 2: CombinatorialIdentity.lean — Vandermonde instance at $n = 7$.

A.3 RamseySearch ($R(3, 3) \geq 6$)

```
import Mathlib.Combinatorics.SimpleGraph.Basic
import Mathlib.Tactic

/-- The cycle C_5 is triangle-free,
    hence R(3,3) >= 6. -/
theorem ramsey_3_3_ge_6 :
    let adj : Fin 5 → Fin 5 → Bool :=
      fun i j => decide (
        (i.val + 1) % 5 = j.val ∨
        (j.val + 1) % 5 = i.val)
    -- No monochromatic triangle in
    -- either colour
    ∀ a b c : Fin 5,
      a ≠ b → b ≠ c → a ≠ c →
      ¬(adj a b = true ∧
        adj b c = true ∧
        adj a c = true) ∨
      ¬(adj a b = false ∧
        adj b c = false ∧
        adj a c = false)   by
  native_decide
```

Listing 3: RamseySearch.lean — $R(3, 3) \geq 6$ via C_5 colouring.

References

[1] K. Appel and W. Haken. Every planar map is four colorable. *Bulletin of the American Mathematical Society*, 82(5):711–712, 1976.

[2] M. Bläser. On the complexity of the multiplication of matrices of small formats. *Journal of Complexity*, 19(1):43–60, 2003.

[3] L. de Moura, S. Kong, J. Avigad, F. van Doorn, and M. von Raumer. The Lean 4 theorem prover and programming language. In *Proceedings of the 28th International Conference on Automated Deduction (CADE-28)*, LNCS 12699, pages 625–635. Springer, 2021.

[4] G. Desaulniers, J. Desrosiers, and M. M. Solomon, editors. *Column Generation*. Springer, New York, 2005.

[5] H. W. Gould. *Combinatorial Identities: A Standardized Set of Tables Listing 500 Binomial Coefficient Summations*. Morgantown, WV, revised edition, 1972.

[6] The Mathlib Community. Mathlib: the math library of Lean 4. `https://github.com/leanprover-community/mathlib4`, 2024.

[7] S. Mattheus and J. Verstraëte. The asymptotics of $r(4, t)$. *Annals of Mathematics*, 199(2):919–941, 2024. DOI: 10.4007/annals.2024.199.2.8.

[8] S. Radziszowski. Small Ramsey numbers. *Electronic Journal of Combinatorics*, Dynamic Survey DS1, Revision 18, 2026.

[9] J. Riordan. *Combinatorial Identities*. John Wiley & Sons, New York, 1968.

[10] V. Strassen. Gaussian elimination is not optimal. *Numerische Mathematik*, 13:354–356, 1969.